

Weak-RF Susceptibility Solver for the Magnetometry Page

Alkali Pumping application

1 Purpose and phase convention

The solver calculates the linear response of a selected observable O in the upper ground hyperfine manifold to a weak, linearly polarized RF field. The RF Hamiltonian is defined as

$$\frac{H_{\text{rf}}(t)}{\hbar} = \Omega_{\text{rf}} F_i \cos(\omega t), \quad (1)$$

where $i \in \{x, y, z\}$ is the selected laboratory RF axis, $\omega = 2\pi f_{\text{rf}}$, and Ω_{rf} is an angular Rabi frequency. The density matrix is expanded directly about zero RF drive:

$$\rho(t) = \rho_0 + \Omega_{\text{rf}} \rho^{(1)}(t) + \mathcal{O}(\Omega_{\text{rf}}^2). \quad (2)$$

Thus the calculation does not assign an arbitrary finite value to Ω_{rf} .

The real cosine drive contains two Fourier components,

$$\cos(\omega t) = \frac{1}{2}e^{-i\omega t} + \frac{1}{2}e^{+i\omega t}. \quad (3)$$

We use $\chi_+(\omega)$ for the coefficient of $e^{-i\omega t}$ in $\partial\langle O \rangle / \partial\Omega_{\text{rf}}$.

2 Basis, drive, and readout

Let $|a\rangle = |F, m\rangle$ denote a ground-state basis vector. The plotted RF observable belongs to the upper ground manifold

$$F = F_+ \equiv I + \frac{1}{2}. \quad (4)$$

The available observables are spin components F_j and the off-diagonal alignment operator

$$Q_{ij} = \frac{1}{2}(F_i F_j + F_j F_i). \quad (5)$$

For the Q_{ij} option, the indices are chosen cyclically from the RF axis:

$$x \rightarrow yz, \quad y \rightarrow zx, \quad z \rightarrow xy. \quad (6)$$

The laboratory operators are expressed in the local frame whose z axis is the selected quantization axis.

For any readout operator O ,

$$\langle O \rangle = \text{Tr}(\rho O) = \sum_{a,b} \rho_{ab} O_{ba}. \quad (7)$$

3 Independent adjacent-transition response

Consider the adjacent pair

$$|a\rangle = |F, m\rangle, \quad |b\rangle = |F, m-1\rangle. \quad (8)$$

Its steady-state population difference is

$$D_m = p_a - p_b. \quad (9)$$

The signed transition angular frequency is

$$\omega_m = 2\pi\nu_{ab}, \quad (10)$$

where ν_{ab} includes the static-field and diagonal light-shift contributions. The adjacent-coherence decay rate is

$$\Gamma_m = \Gamma_{\text{OP},m} + \Gamma_{\text{ER},m} + \Gamma_{\text{SE},m}. \quad (11)$$

Define the RF drive matrix elements

$$d_{ab} = \langle a|F_i|b\rangle, \quad d_{ba} = \langle b|F_i|a\rangle. \quad (12)$$

The coefficients of $e^{-i\omega t}$ in the two ordered coherences, per unit Ω_{rf} , are

$$\frac{\partial \rho_{ab,+}}{\partial \Omega_{\text{rf}}} = \frac{id_{ab}D_m/2}{\Gamma_m + i(\omega_m - \omega)}, \quad (13)$$

$$\frac{\partial \rho_{ba,+}}{\partial \Omega_{\text{rf}}} = \frac{-id_{ba}D_m/2}{\Gamma_m + i(-\omega_m - \omega)}. \quad (14)$$

The first denominator is the co-rotating resonance and the second is the counter-rotating contribution. Both are retained because the applied field is a real cosine. In particular, the second term correctly represents a negative signed transition frequency that resonates at positive drive frequency.

Using Eq. (7), the total complex susceptibility is

$$\chi_+(\omega) = \sum_m \left[O_{ba} \frac{\partial \rho_{ab,+}}{\partial \Omega_{\text{rf}}} + O_{ab} \frac{\partial \rho_{ba,+}}{\partial \Omega_{\text{rf}}} \right]. \quad (15)$$

Only adjacent coherences within F_+ that have nonzero drive and readout matrix elements contribute.

4 Full coherence-space solver

The coupled solver retains every ordered off-diagonal coherence. Write

$$\mathbf{c} = (\rho_{a_1 b_1}, \rho_{a_2 b_2}, \dots)^T, \quad a_k \neq b_k. \quad (16)$$

Its linearized equation of motion is

$$\frac{d\mathbf{c}}{dt} = G\mathbf{c} + \Omega_{\text{rf}} (\mathbf{s}e^{-i\omega t} + \mathbf{s}_-e^{+i\omega t}). \quad (17)$$

For the $e^{-i\omega t}$ component, set

$$\mathbf{c}(t) = \Omega_{\text{rf}} \mathbf{c}_+(\omega) e^{-i\omega t} + \dots. \quad (18)$$

Substitution into Eq. (17) gives

$$(-i\omega I - G)\mathbf{c}_+ = \mathbf{s}, \quad (19)$$

and therefore

$$\boxed{\mathbf{c}_+(\omega) = (-i\omega I - G)^{-1} \mathbf{s}.} \quad (20)$$

For an ordered coherence (a, b) , the RF source and observable readout are

$$s_{ab} = \frac{i}{2}(F_i)_{ab}(p_a - p_b), \quad (21)$$

$$r_{ab} = O_{ba}. \quad (22)$$

Combining Eqs. (20) and (22) yields the implemented resolvent expression

$$\boxed{\chi_+(\omega) = \mathbf{r}^T(-i\omega I - G)^{-1} \mathbf{s}.} \quad (23)$$

Here T means the algebraic row-vector contraction used by the density-matrix readout. The necessary complex conjugate relationships are already present in the Hermitian operator matrix elements and in the ordered-coherence basis.

4.1 Local generator

For one alkali species, let

$$\nu_a = \nu_{\text{LS},a} + \nu_{B,a} \quad (24)$$

be the light-shift plus magnetic frequency assigned to state a . The bare coherence generator for (a, b) contains

$$G_{ab,ab}^{(0)} = -\frac{\Gamma_{\text{OP},a} + \Gamma_{\text{OP},b}}{2} - i, 2\pi(\nu_a - \nu_b). \quad (25)$$

Electron randomization and spin exchange are represented by linear coherence maps M_{ER} , $M_{\text{SE,self}}$, and $M_{\text{SE,cross}}$. The local generator is

$$\begin{aligned} G_{\text{loc}} = G^{(0)} &+ R_{\text{ER}}(M_{\text{ER}} - I) + R_{\text{SE,self}}(M_{\text{SE,self}} - I) \\ &+ R_{\text{SE,cross}}(M_{\text{SE,cross}} - I). \end{aligned} \quad (26)$$

These maps are the derivatives of the corresponding density-matrix collision maps about the calculated steady state. Thus their off-diagonal entries can transfer perturbations between different coherences rather than merely adding a scalar linewidth.

4.2 Two-species block generator

For active alkalis A and B, concatenate their ordered coherences:

$$\mathbf{c} = \begin{pmatrix} \mathbf{c}_A \\ \mathbf{c}_B \end{pmatrix}. \quad (27)$$

The joint generator is

$$G = \begin{pmatrix} G_A & R_{\text{SE,cross},A}M_{A \leftarrow B} \\ R_{\text{SE,cross},B}M_{B \leftarrow A} & G_B \end{pmatrix}. \quad (28)$$

For the Alkali-A curve, only A is directly driven and read out:

$$\mathbf{s}^{(A)} = \begin{pmatrix} \mathbf{s}_A \\ 0 \end{pmatrix}, \quad \mathbf{r}^{(A)} = \begin{pmatrix} \mathbf{r}_A \\ 0 \end{pmatrix}. \quad (29)$$

The B coherences can nevertheless modify the A response through the two off-diagonal spin-exchange blocks. The B curve is defined conversely.

5 In-phase, quadrature, and amplitude

The real time-domain susceptibility is defined with the RF cosine as its phase reference:

$$\frac{\partial \langle O(t) \rangle}{\partial \Omega_{\text{rf}}} = X(\omega) \cos(\omega t) + Y(\omega) \sin(\omega t). \quad (30)$$

Since χ_+ is the coefficient of $e^{-i\omega t}$,

$$X(\omega) = 2 \operatorname{Re} \chi_+(\omega), \quad (31)$$

$$Y(\omega) = 2 \operatorname{Im} \chi_+(\omega), \quad (32)$$

$$A(\omega) = \sqrt{X(\omega)^2 + Y(\omega)^2} = 2|\chi_+(\omega)|. \quad (33)$$

Thus X is the in-phase component, Y is the quadrature component, and A is the nonnegative amplitude. No sign reversal is applied internally.

The optional *Add π* controls are display transformations only:

$$X_{\text{plot}} = -X \quad \text{or} \quad Y_{\text{plot}} = -Y. \quad (34)$$

They do not change χ_+ , the generator, or the amplitude.

6 Units and optional display factors

The spin and alignment operators are represented in dimensionless angular momentum units. Because the response is differentiated with respect to the angular Rabi frequency Ω_{rf} in radians per second, the unscaled susceptibility has the corresponding inverse-frequency unit (seconds for a dimensionless spin observable).

Two optional display factors may be applied after solving:

$$(A, X, Y)_{\text{plot}} = C(A, X, Y), \quad C = \begin{cases} 1, & \text{no optional factor,} \\ \Gamma_{\text{ref}}, & \text{relaxation normalized,} \\ n, & \text{density factored,} \\ \Gamma_{\text{ref}} n, & \text{both selected.} \end{cases} \quad (35)$$

Here Γ_{ref} is the adjacent-coherence relaxation rate selected by the application's reference rule and n is the alkali number density. These factors do not alter the underlying linear-response solution.

7 Computational form

The implementation diagonalizes G once,

$$G = V \Lambda V^{-1}, \quad (36)$$

and evaluates the response at every requested frequency as

$$\chi_+(\omega) = \sum_k \frac{(\mathbf{r}^T V)_k (V^{-1} \mathbf{s})_k}{-i\omega - \lambda_k}. \quad (37)$$

A typical eigenvalue is $\lambda_k = -\Gamma_k - i\omega_k$. Equation (37) makes the physical structure clear: each coherence mode contributes a complex Lorentzian centered near its precession frequency and weighted by how strongly the RF drive excites it and the selected observable detects it.

8 Summary of the solver

1. Calculate the zero-RF steady-state populations and diagonal state frequencies.
2. Construct all ordered coherences and the generator G , including optical relaxation, electron randomization, and spin exchange.
3. Construct the source vector $\mathbf{s}$ from RF matrix elements and population differences.
4. Construct the readout vector $\mathbf{r}$ from the chosen F_j or Q_{ij} observable.
5. Evaluate $\chi_+(\omega) = \mathbf{r}^T(-i\omega I - G)^{-1}\mathbf{s}$.
6. Convert the phasor to $X = 2\Re\chi_+$, $Y = 2\Im\chi_+$, and $A = 2|\chi_+|$.
7. Apply only the user-selected display factors and optional π phase flips.